

KNOXVILLE 69-13.2kV RELAY EQUIPMENT MATERIAL LIST									
		QUANTITY / DEVICE NAMES							
ITEM NO.	EQUIPMENT DESCRIPTION	TOTAL	PNL. 0 RTU	PNL. 1 POLO 69kV LINE & 69kV BUS	PNL. 2 MISSOURI CITY 69kV LINE	PNL. 3 SUMMERSET 69kV LINE	PNL. 4 AVALON 69kV LINE	TERM CAB.	Loose and Not Mounted
1R	Distance relay Mounting: Horizontal Rack mount, 3 RU Power supply voltage: 48/125V DC or 120V AC Control voltage: 48V DC Current input rating: 5A Voltage input rating: 300V AC Standard I/O: 6 optoisolated inputs, 3 high current outputs, 5 outputs Additional I/O: 8 inputs, 12 high current outputs User interface: Safelock trip/close pushbuttons with configurable labels Comm. interface: 2 EIA-232 DB9, 1 EIA-485 terminal block, 2 10/100Base-T, 1 BNC IRIG, 1 USB B Communications: Standard SEL 0311C11HR3E5261 See catalog cut	4	0	1 21P-P	1 21P-M	1 21P-S	1 21P-A	0	0

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1T	Distance relay Mounting: Horizontal Rack mount, 3 RU Power supply voltage: 48/125V DC or 120V AC Control voltage: 48V DC Current input rating: 5A Voltage input rating: 300V AC Standard I/O: 6 optoisolated inputs, 3 high current outputs, 5 outputs Additional I/O: 8 inputs, 12 high current outputs User interface: Hard-wired pushbuttons Comm. interface: 2 EIA-232 DB9, 1 EIA-485 terminal block, 2 10/100Base-T, 1 BNC IRIG, 1 USB B Communications: Standard SEL 0311C11HH3E5261 See catalog cut	1	0	1 21-69B	0	0	0	0	0

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ITEM NO.	EQUIPMENT DESCRIPTION	TOTAL	PNL. 0 RTU	PNL. 1 POLO 69kV LINE & 69kV BUS	PNL. 2 MISSOURI CITY 69kV LINE	PNL. 3 SUMMERSET 69kV LINE	PNL. 4 AVALON 69kV LINE	TERM CAB.	Loose and Not Mounted
2C	Line current differential relay Mounting: Horizontal Rack mount, 4 RU Power supply voltage: 48/125V DC or 125V AC Control voltage: 48V DC Current input rating: 5A Voltage input rating: 150V AC Standard I/O: 6 optoisolated inputs, 8 outputs, 6 high speed - high current outputs Additional I/O: 8 optoisolated inputs, 12 high current outputs User interface: Hard-wired pushbuttons Comm. interface: 2 EIA-232 DB9, 1 EIA-485 terminal block, 2 10/100Base-T, 2 Fiber optic SM ST 1300 nm, 1 IRIG terminal block Communications: Standard serial protocols, serial DNP3 with two 10/100BASE-T ethernet ports, SEL ASCII via Telnet SEL 0311L1JDD325262XX See catalog cut	4	0	1 21B-P	1 21B-M	1 21B-S	1 21B-A	0	0
11B	Gas tube coaxial surge protector 200-2005 and mounting bracket 200-2006 Comm. interface: 2 TNC SEL 915900139 See catalog cut	1	0	0	0	0	0	1	0
11D	BNC terminator, 50Ω SEL 240-1800 See catalog cut	2	1	0	0	0	0	0	1
11G	GPS GNSS antenna for satellite clock SEL 9524A See catalog cut	1	0	0	0	0	0	0	1

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11J	Satellite clock with blank bracket on right side Mounting: Horizontal Rack mount, 1 RU Power supply voltage: 18-300V DC or 85-264V AC Comm. interface: 1 EIA-232 DB9, 7 BNC IRIG, 1 TNC SEL 24070000B See catalog cut	1	1 CLOCK	0	0	0	0	0	0
12E	Revenue meter Power supply voltage: 24/48V DC Control voltage: 48V DC Current input rating: 22A Voltage input rating: 300V AC Standard I/O: 2 inputs, 3 outputs Additional I/O: 4 inputs, 4 outputs Comm. interface: 2 EIA-232 DB9, 2 EIA-485 terminal block, 1 10/100Base-T, 1 IRIG terminal blocks, 1 Telephone modem, 1 Optical Communications: SEL ASCII, SEL LMD, SEL Fast Meter, SEL Compressed ASCII, MV-90 Translation™, Modbus® RTU/TCP, MIRRORRED BITS® Communications, Telnet, DNP3 Level 2 Outstation and LAN/WAN(3), and Synchrophasors(3); Standard Communications Protocols Meter form: Form 9 (4-wire wye; 3 PTs, 3 CTs) SEL 0735VX00922EGXA2XX16101XX See catalog cut	1	0	0	1 RM-T1	0	0	0	0

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13A	Port server Mounting: Horizontal Rack mount, 1 RU Power supply voltage: 48/125V DC, 120V AC Standard I/O: 1 input, 1 output Comm. interface: 16 EIA-232/EIA-485 DB9, 1 EIA-232/EIA-485 terminal block, 2 10/100Base-T, 1 BNC IRIG In, 1 BNC IRIG Out, 1 IRIG terminal block, 1 USB A, 1 USB B SEL 3610XHB0XXX0 See catalog cut	1	1 RTU1A	0	0	0	0	0	0
13N	Real-time automation controller (RTAC) Operating system: Real-time automation controller (RTAC) Software: Report Generator/Email Client, FileIO RAM: 8 GB SSD Slot 1: pSLC 120 GB Industrial Grade SSD Mounting: Horizontal Rack mount, 3 RU Power supply voltage: 2 supplies, both at 48/125V DC or 120V AC Standard I/O: 1 input, 1 output Comm. interface: 16 EIA-232/EIA-485 RJ45, 4 10/100/1000Base-T, 1 front 10/100/1000Base-T, 4 USB 2.0 Type A, 2 front USB 3.1 Type A, 1 BNC IRIG In, 1 BNC IRIG Out, 1 DisplayPort 1.2 SEL 3350#HV5J See catalog cut	1	1 RTAC	0	0	0	0	0	0
14A	Axion chassis/backplane 10 slot Mounting: Horizontal Rack mount, 5 RU SEL 2242R1X0 See catalog cut	1	1 RTU	0	0	0	0	0	0

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14C	Axion power module and EtherCAT cable, shielded, with RJ45 connector at each end Power supply voltage: 24/48V DC Comm. interface: 2 10/100Base-T for EtherCAT Length: 3 feet SEL 224321X0, C627F-003 See catalog cut	1	1	0	0	0	0	0	0
14D	Wetting voltage kit Twenty four - 4 prong jumpers For use with Axion digital input module SEL 915900240 See catalog cut	1	1	0	0	0	0	0	0
14E	Axion digital input module Control voltage: 48V DC Standard I/O: 24 inputs SEL 22442222X0 See catalog cut	3	3	0	0	0	0	0	0
14F	Axion digital output module, 16 form A outputs SEL 22443131X0 See catalog cut	1	1	0	0	0	0	0	0
14G	Axion digital output module, 10 high speed - high current outputs SEL 22445151X0 See catalog cut	1	1	0	0	0	0	0	0
14J	Axion power module and EtherCAT cable, shielded, with RJ45 connector at each end Power supply voltage: 125/250V DC, 120/240V AC Comm. interface: 2 10/100Base-T for EtherCAT Length: 3 feet SEL 224311X0, C627F-003 See catalog cut	1	1	0	0	0	0	0	0

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14R	Axion RTAC module, 1 output Comm. interface: 4 EIA-232/EIA-485 DB9, 2 10/100Base-T, 1 BNC IRIG In, 1 BNC IRIG Out, 1 USB B Communications: Client: SEL, DNP3, Modbus, IEC 60870-5-101/104, C37.118 Synchrophasors, L&G; 8979, CP2179, SNMP Server: SEL, DNP3, Modbus, L&G; 8979, SES-92, IEC 60870-5-101/104, C37.118 Synchrophasors, CDC Type II Peer-to-Peer: Mirrored Bits SEL 2241#JD94 See catalog cut	1	1	0	0	0	0	0	0
35B	Lock-out relay Handle style: Oval Coil rating: 48V DC Number of decks: 8 Electroswitch Series 24 7828CC See catalog cut	1	0	1 86-69B	0	0	0	0	0
37	Kirk Key interlock, consisting of one lock, operating a single shaft which operates two make and two break contacts, SD series, each with separate keying, marked as shown in table, reversed for projection mounting of cylinder Kirk Key KFL400010S - - - R See catalog cut	5	0	1 43-BP-P Mark Key: K-P	1 43-BP-M Mark Key: K-M	1 43-BP-S Mark Key: K-S	1 43-BP-A Mark Key: K-A	0	1 Mark Key: K- _

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37B	Kirk key interlock cylinder, Type B, base mounted, no auxiliary switch, 1" bolt extension when key removed, each keyed to corresponding interlock Item 37, key and lock to be marked as shown in table, with 3/8"-16 x 3/4" tamperproof mounting hardware, and dust cover Kirk Key KBL010010S - CT - - See catalog cut	5	0	0	0	0	0	0	5 Mark Keys: K-P K-M K-S K-A K-__
37C	3/8" tamperproof bolt key for installing Kirk Key interlock cylinder at switch, to be left in control building after installation Kirk Key 21HKHKA See catalog cut	2	0	0	0	0	0	0	2
41	Test switch, 30-pole Mounting: Rack mount, 3 RU Position A: C-C,C-C,C-C,C-C,P,P Position B: P,P,P,P,P,P,P,P,P Position C: P,P,P,P,P,P,P,P,P Cover: Full length, clear ABB Type FT-19R FRXG 083 001 001 See catalog cut	9	0	3 TS1-21-69B TS1-21P-P TS1-21B-P	2 TS1-21P-M TS1-21B-M	2 TS1-21P-S TS1-21B-S	2 TS1-21P-A TS1-21B-A	0	0
42	Test switch, 10-pole Arrangement: P,C-C,P,C-C,P,C-C,P Cover: Black ABB Type FT-1 129A528G01 See catalog cut	6	0	1 TS-MM-P	3 TS-MM-M TS-MM-T1 TS-RM-T1	1 TS-MM-S	1 TS-MM-A	0	0
44	Test switch, 30-pole Mounting: Rack mount, 3 RU Position A: P,P,P,P,P,P,P,P,P Position B: P,P,P,P,P,P,P,P,P Position C: P,P,P,P,P,P,P,P,P Cover: Full length, clear ABB Type FT-19R FRXG 001 001 001 See catalog cut	13	0	4 TS2-21-69B TS2-21P-P TS3-21P-P TS2-21B-P	3 TS2-21P-M TS3-21P-M TS2-21B-M	3 TS2-21P-S TS3-21P-S TS2-21B-S	3 TS2-21P-A TS3-21P-A TS2-21B-A	0	0

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44A	Test switch, 14-pole Arrangement: P,P,P,P,P,P,P,P,P,P,P,P Cover: Clear ABB Type FT-14 FT4A14T14CN4001 See catalog cut	1	0	1 TS-86-69B	0	0	0	0	0
50	Meter which displays, stores and transmits: 3-phase volts, 3-phase amperes, watts, VARs, kVA, power factor, neutral amperes, kWh and kVARh in/out Power supply voltage: 48-250V DC or 55-240V AC, with monitoring Current input rating: 5A Voltage input rating: 120V AC Comm. interface: 1 EIA-232/EIA-485 terminal block, 1 service-only RJ45 Communications: DNP with RS-485 Phases: 3 Wires: 4 Bitronics M650M3P510 See catalog cut	5	0	1 MM-P	2 MM-M MM-T1	1 MM-S	1 MM-A	0	0
76	Indicating lamp, round type, complete with resistor and LED lamp, Type ET-16 Lamp voltage: 48V DC Lamp color: Amber Lens color: Clear GE 116B6708G42A52C4 See catalog cut	10	0	4	2	2	2	0	0
90C	9-pin male D-sub connector with screw terminals and plastic backshell Wire size: 18 AWG max Amphenol D-ST L717DE09PST1 See catalog cut	0	0	0	0	0	0	0	0
92	Resistor, 1/4 Watt, 150Ω	2	0	0	1	0	1	0	0

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99	Service to change jumper on internal circuit board of microprocessor relay by soldering or other appropriate means per manufacturer instructions See catalog cut	4	0	1 for 21P-P	1 for 21P-M	1 for 21P-S	1 for 21P-A	0	0
100	Terminal block, 2-point, 600 V, 175A, #14 to 2/0 lug Square D 9080LBA262101 See catalog cut	0	0	0	0	0	0	0	0
101A	Configurable label set for microprocessor relay, printed and installed per drawings	4	0	1	1	1	1	0	0
101B	Nameplates, black-on-white (black with white letters), engraved plastic per schedule	123	0	42	31	25	25	0	0
101C	Nameplates, white-on-black (white with black letters), engraved plastic per schedule	32	0	8	8	8	8	0	0
102	Ground bus, 1" x 1/4" copper, horizontally at bottom	4	0	1	1	1	1	0	0
102A	4/0 copper stranded to 1" x 1/4" copper bar connector	2	1	0	0	0	0	1	0
102C	Coupling bar, 1" x 1/4" copper, for connecting copper ground bus from one relay panel to another	4	1	1	1	1	0	0	0
103	Terminal block, 12 point, 10-32 washer head screws, with white marking strip GE EB25B12 See catalog cut	92	0	14	16	16	16	25	5
104	Terminal block, shorting type, 6 point, 10-32 washer head screws GE EB27B06S See catalog cut	50	0	8	4	4	4	25	5

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105C	Terminal, one piece, seamless barrel, tinned copper, boxed ring tongue, sized for the appropriate conductor Burndy YAV, or equivalent See catalog cut	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	0
106	Switchboard wire, tinned #12 AWG, 65-strand, 30 mils of cross-linked polyethylene insulation, Underwriters Lab type SIS, flame resistance VW-1 rating, gray color, 600V, 90°C	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	0
107B	Shielded pairs, #18 AWG, for low level signals	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	0
108	1-pole fuse block, 250V, 30A, screw terminals, spring reinforced clips for Class H non-renewable fuses Mersen 20316 See catalog cut	8	4	0	1	0	0	0	3
109	2-pole fuse block, 250V, 30A, screw terminals, spring reinforced clips for Class H non-renewable fuses Mersen 20317 See catalog cut	22	5	4	3	3	2	0	5
110	3-pole fuse block, 250V, 30A, screw terminals, spring reinforced clips for Class H non-renewable fuses Mersen 20318 See catalog cut	22	0	6	5	3	3	0	5
111	Fuse, 250V, 30A Eaton Bussman FRN-R-30 See catalog cut	20	0	8	0	0	0	0	12
111A	Fuse, 250V, 20A Eaton Bussman FRN-R-20 See catalog cut	20	2	2	2	2	2	0	10

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111C	Fuse, 250V, 6A Eaton Bussman FRN-R-6 See catalog cut	90	12	16	20	13	11	0	18
112	Terminal block points, captive screw for ring tongue terminals, mounted in DIN rail terminal block system, complete with DIN rail, end caps, and spacers as required Color: Blue Phoenix RT 5 BU 3049123 See catalog cut	132	120	0	0	0	0	0	12
112A	Terminal block points, captive screw for ring tongue terminals, mounted in DIN rail terminal block system, complete with DIN rail, end caps, and spacers as required Color: Gray Phoenix RT 5 3049026 See catalog cut	132	120	0	0	0	0	0	12
118	Flip turn door handle, size as required Finish: Polished chrome Southco 62-__-1 See catalog cut	10	2	2	2	2	2	0	0
121	Removable lifting eye capable of lifting shipping section	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	0
123	Fuse puller Bussman FP-2 See catalog cut	2	0	0	0	0	0	0	2
129	Panduit, size as required	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	0	0
130A	Stainless steel door handle assembly with 3 point latch per drawings	1	0	0	0	0	0	1	0
140	Wire markers, all conductors used for internal wiring shall be tagged with the wire name and destination at both ends	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	1 Lot	0

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141	Touch-up paint from same lot used to paint panel exterior 1 pint	1	0	0	0	0	0	0	1
150	Duplex receptacle, 20A, metal face plate, flush mount, not GFCI, in metal outlet box	3	1	0	1	0	1	0	0
150A	4-plex receptacle, 20A, metal face plates, in metal box, not GFCI	1	1	0	0	0	0	0	0
161	Blank plate with returns Mounting: Rack mount, 1 RU	2	2	0	0	0	0	0	0
162	Blank plate with returns Mounting: Rack mount, 2 RU	2	1	0	1	0	0	0	0
163	Blank plate with returns Mounting: Rack mount, 3 RU	21	6	1	2	6	6	0	0
163A	Blank plate with returns and cutouts Mounting: Rack mount, 3 RU	4	0	1	1	1	1	0	0
163B	Subpanel Mounting: Rack mount, 3 RU	1	1	0	0	0	0	0	0
164	Blank plate with returns Mounting: Rack mount, 4 RU	3	3	0	0	0	0	0	0
164A	Blank plate with returns and cutouts Mounting: Rack mount, 4 RU	5	0	1	2	1	1	0	0
166A	Blank plate with returns and cutouts Mounting: Rack mount, 6 RU	2	0	1	1	0	0	0	0
202C	RG-58A/U coax cable, male BNC connectors on both ends, 50Ω Length: 5 feet SEL C953-005	2	0	2	0	0	0	0	0
203E	RG-8X, TNC male to TNC male cable, 50Ω Length: 10 feet SEL C965-010	1	0	0	0	0	0	0	1

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203G	RG-8X, TNC male to TNC male cable, 50Ω Length: 20 feet SEL C965-020	1	0	0	0	0	0	0	1
215G	EIA-232 serial cable with IRIG, standard-shielded, male RJ-45 to male DB9 Length: 20 feet SEL C489-020	3	0	0	0	0	0	0	3
215H	EIA-232 serial cable with IRIG, standard-shielded, male RJ-45 to male DB9 Length: 25 feet SEL C489-025	5	0	0	0	0	0	0	5
215J	EIA-232 serial cable with IRIG, standard-shielded, male RJ-45 to male DB9 Length: 30 feet SEL C489-030	2	0	0	0	0	0	0	2

KNOXVILLE 69-13.2KV RELAY EQUIPMENT CABLE LIST

ITEM NO	CABLE TYPE	CABLE LENGTH	FROM				TO			
			PANEL	DEVICE NAME	DEVICE TYPE	PORT	PANEL	DEVICE NAME	DEVICE TYPE	PORT
215G	C489	20	PNL. 1	21P-P	SEL-311C	2	PNL. 0	RTAC	SEL-3350	1
215G	C489	20	PNL. 1	21B-P	SEL-311L	2	PNL. 0	RTAC	SEL-3350	2
					MODEM		PNL. 0	RTAC	SEL-3350	3
215H	C489	25	PNL. 2	21P-M	SEL-311C	2	PNL. 0	RTAC	SEL-3350	4
215H	C489	25	PNL. 2	21B-M	SEL-311L	2	PNL. 0	RTAC	SEL-3350	5
215H	C489	25	PNL. 3	21P-S	SEL-311C	2	PNL. 0	RTAC	SEL-3350	6
215H	C489	25	PNL. 3	21B-S	SEL-311L	2	PNL. 0	RTAC	SEL-3350	7
215J	C489	30	PNL. 4	21P-A	SEL-311C	2	PNL. 0	RTAC	SEL-3350	8
215J	C489	30	PNL. 4	21B-A	SEL-311L	2	PNL. 0	RTAC	SEL-3350	9
215G	C489	20	PNL. 1	21-69B	SEL-311C	2	PNL. 0	RTAC	SEL-3350	10
215H	C489	25	PNL. 2	RM-T1	SEL-735	2	PNL. 0	RTAC	SEL-3350	11
203E	C965	10		ANT	SEL-9524A		TERM CAB.	SURGE PRT.	SEL	
203G	C965	20	TERM CAB.	SURGE PRT.	SEL		PNL. 0	CLOCK	SEL-2407	ANT
202C	C953	5	PNL. 0	CLOCK	SEL-2407	ANT	PNL. 0	RTAC	SEL-3350	IRIG-IN
202C	C953	5	PNL. 0	RTU	SEL-2241	A-IRIG-IN	PNL. 0	RTAC	SEL-3350	IRIG-OUT
14C	224321X0, C627F	3	PNL. 0	RTU	SEL-2241	A-ETH1	PNL. 0	RTU1A	SEL-3610	ETH1

DATE	USER	DESCRIPTION
2025-08-22	JPG	Initial
2025-08-27	JPG	Rev 1